

WEXFORD COUNTY AP, MI, USA

WMO: 726384

Lat: 44.283N

Lon: 85.417W

Elev: 1305

StdP: 14.02

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-0.4	-13.1	2.8	-5.9	-6.6	4.1	0.4	23.3	23.2	20.7	23.4	2.2	250	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	87.7	71.0	83.8	68.7	81.5	67.4	74.4	82.5	72.4	80.1	70.6	77.3	10.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	125.1	78.7	70.2	116.5	76.5	68.3	108.9	74.8	38.8	82.5	37.0	79.3	35.2	77.0	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.6	16.7	-17.4	92.3	8.5	3.5	-23.5	94.8	-28.5	96.8	-33.3	98.8	-39.5	101.3
			-16.9	77.7	7.6	3.2	-22.4	80.0	-26.9	81.8	-31.2	83.6	-36.7	85.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	19.4	20.5	29.4	41.9	53.8	63.3	67.4	65.6	58.7	46.9	35.2	25.4
	DBStd	19.35	10.92	11.32	11.78	9.49	8.89	7.35	6.41	5.96	8.15	8.69	8.97	9.51
	HDD50	4163	950	825	649	278	60	3	0	0	16	169	451	762
	HDD65	7961	1415	1245	1105	695	364	116	48	65	222	565	895	1226
	CDD50	2001	0	0	9	34	178	402	540	483	276	73	6	0
	CDD65	326	0	0	0	1	17	66	123	84	32	3	0	0
	CDH74	3609	0	0	7	22	248	792	1357	828	326	29	0	0
	CDH80	887	0	0	1	1	50	212	378	177	66	2	0	0
(14) Wind	WSAvg	7.2	8.0	8.0	7.6	8.4	7.6	6.5	5.9	5.6	6.0	7.3	7.8	7.8
(15) Precipitation	PrecAvg	34.1	2.1	1.5	1.9	3.0	3.3	3.5	3.2	3.3	3.1	3.5	2.5	2.1
	PrecMax	41.3	4.0	3.2	4.1	4.7	6.8	9.7	9.2	7.3	6.1	9.2	5.4	5.3
	PrecMin	26.5	0.7	0.6	0.6	0.9	0.6	1.3	1.3	1.4	0.8	0.8	1.0	0.8
	PrecStd	4.4	0.9	0.7	0.9	1.1	1.7	2.0	1.9	1.4	1.2	1.9	1.2	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.6	52.3	71.5	77.3	85.2	90.4	92.0	89.9	86.8	78.7	64.1	54.4
		MCWB	44.8	47.2	60.4	59.4	67.5	72.8	74.7	71.6	69.7	64.3	54.8	51.2
	2%	DB	41.2	44.9	59.4	70.5	80.8	85.3	88.1	84.2	81.4	72.3	57.3	46.0
		MCWB	38.2	40.0	49.3	55.4	64.7	68.6	72.2	69.4	66.4	61.7	51.9	43.2
	5%	DB	36.5	39.1	52.7	64.3	76.5	81.8	83.7	81.4	77.7	67.5	54.3	41.2
		MCWB	35.0	35.5	46.3	50.8	61.7	67.0	69.3	68.0	66.1	59.9	49.2	38.7
	10%	DB	34.2	35.9	46.0	59.4	71.9	78.8	81.3	78.9	73.7	62.4	49.9	37.0
		MCWB	33.1	33.5	40.2	48.7	59.4	65.5	68.0	66.6	64.1	56.0	45.5	35.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	47.1	61.1	62.7	71.0	75.1	78.4	76.0	73.5	69.0	57.1	51.5
		MCDB	46.8	49.2	70.0	72.5	80.1	85.6	88.3	81.8	81.2	74.5	61.1	53.8
	2%	WB	38.8	40.3	53.4	57.7	67.5	72.0	75.1	73.3	70.5	64.1	53.3	43.9
		MCDB	40.8	44.4	57.5	67.1	76.0	81.3	83.4	80.4	75.9	69.2	57.3	46.0
	5%	WB	35.2	36.1	46.1	53.9	64.4	69.9	73.0	71.3	68.4	60.2	49.6	38.8
		MCDB	36.2	38.8	50.8	62.0	72.9	77.5	80.5	78.0	74.5	66.2	53.7	40.9
	10%	WB	32.8	33.2	40.7	49.8	61.2	67.9	70.8	69.4	66.0	56.5	45.0	35.3
		MCDB	34.1	35.6	46.0	58.5	68.9	75.3	77.7	74.8	73.1	61.5	48.9	37.0
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	16.5	19.4	22.1	24.2	23.6	24.1	23.0	22.9	18.8	14.0	11.7
		MCDBR	14.0	17.5	25.6	31.4	30.0	27.7	27.3	26.2	26.2	26.4	22.1	16.1
		MCWBR	12.7	14.0	17.5	18.7	16.2	14.1	13.4	13.7	14.5	16.7	16.7	13.6
	5% WB	MCDBR	13.1	15.6	23.2	26.6	24.1	22.0	21.9	20.6	20.5	22.8	20.2	14.9
		MCWBR	12.1	13.1	17.4	18.4	15.0	13.1	12.5	12.4	13.1	16.6	16.6	13.4
(40) Clear-Sky Solar Irradiance	taub		0.274	0.291	0.315	0.343	0.372	0.391	0.388	0.377	0.361	0.331	0.297	0.273
	taud		2.358	2.303	2.405	2.372	2.347	2.320	2.333	2.362	2.405	2.490	2.518	2.420
	Ebn at Noon		271	285	293	292	285	279	278	277	273	269	262	260
	Edh at Noon		24	31	32	36	38	40	39	36	32	26	21	21
(44) All-Sky Solar Radiation	RadAvg		393	682	1130	1434	1731	1905	1923	1637	1295	774	442	298
	RadStd		34	74	130	137	158	128	97	106	124	95	62	34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page